

NHAN TRUONG

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EDUCATION

University of Cincinnati, Cincinnati, OH	Aug 2022 - May 2027 (expected)
• Bachelor of Science in Aerospace Engineering	Cumulative GPA: 3.97/4.0
• Relevant courses: Gas Dynamics, Aerodynamics, Rocket Propulsion, Air Breathing Propulsion, Introduction to CFD, Advanced Compressible Flows	
• Project: “ <i>Effect of Embedded Cavities on Fragmentation Behaviors of Droplets in Shock-Laden Flows</i> ”	
• Advisor: Prashant Khare	

RESEARCH INTERESTS

- Shock-Droplet and Shock-Bubble Interactions in High-Speed Flows
- Spray Combustion and Flame Stabilization in High-Speed Propulsion Systems
- Large Eddy Simulation of Turbulent Reacting Flows
- Atomization of Gel Propellant
- Machine Learning-accelerated CFD

RESEARCH AND ACADEMIC EXPERIENCE

Undergraduate Research Assistant	Jan 2024 - Present
University of Cincinnati	Cincinnati, Ohio
1. <u>Learned WENO Reconstruction for Compressible Turbulence</u>	
• Advisor: Prashant Khare	
• Constructing a solver-in-the-loop ML pipeline in JAX-Fluids that learns optimal WENO3 reconstruction weights for 2D compressible turbulence using a differentiable Euler solver	
• Training a JAX neural network on spectrally filtered DNS data with a multi-component loss combining MSE, spectral energy, and weight regularization	
• Benchmarking learned WENO3-NN against WENO3-JS, WENO5-Z, and ALDM using energy spectra, TKE decay, and vorticity correlation with DNS	
2. <u>Gel Atomization in Impinging Jets</u>	
• Advisor: Prashant Khare	
• Implementing Herschel-Bulkley viscosity models in three open-source solvers: JAX-Fluids, MFC, and FluidX3D, covering both compressible Navier-Stokes and lattice-Boltzmann frameworks	
• Validating implementations against analytical solutions and literature benchmarks for 2D Poiseuille flow and lid-driven cavity across Reynolds 100-5000, spanning shear-thickening and shear-thinning fluids	
• Designing and implementing an IMEX ARS (2,2,2) defect-correction time integrator in MFC (Fortran), bypassing the viscous CFL limit for high-viscosity-range fluid while preserving full Navier-Stokes accuracy	
• Conducting impinging jet atomization simulations in MFC to characterize sheet formation, primary breakup mechanisms, and droplet size distributions under variation of ambient pressure	
3. <u>Large Eddy Simulations of Recirculation Structures in Isothermal Swirling Coaxial Jet</u>	
• Advisor: Kamin Manu	
• Examining LES velocity field data of three swirler configurations using ParaView and Python to characterize recirculation zone transitions from pre-vortex breakdown to a central toroidal recirculation zone	
• Calculating swirl number, circulation, and swirl momentum factor to quantify angular momentum penetration into recirculation zone cores	
• Computing mean velocity profiles and turbulence statistics, including TKE and Reynolds stress, to evaluate mixing characteristics and recirculation zone boundaries	

- Implementing POD and DMD in Python to extract dominant spatial modes, temporal coefficients, and mode frequencies for characterizing flow instabilities
4. Effect of Embedded Cavities on Fragmentation Behaviors of Droplets in Shock-Laden Flows
 - Advisor: [Prashant Khare](#)
 - Reviewed the literature on shock-droplet and shock-bubble interactions and identified a research gap around cavity-embedded droplet behavior in supersonic flows
 - Developed and validated numerical framework in STAR-CCM+ by benchmarking against five experimental datasets, analytical solutions, and prior computational studies
 - Executed simulations and analyzed Schlieren visualizations, baroclinic torque distributions, and pressure field evolution in Python to study how the embedded cavity drives droplet deformation and fragmentation
 5. Large Eddy Simulations of Spray and Combustion Processes in a Bluff Body Stabilized High-Speed Combustor
 - Advisors: [Prashant Khare](#), [Kamin Manu](#)
 - Processed LES data using ParaView and Python to evaluate fuel-air mixing, equivalence ratio, heat release rate, temperature, and CO/CO₂ mass fractions at multiple axial stations downstream of V-gutter
 - Applied FFT analysis to velocity and pressure time series to identify vortex shedding frequencies and longitudinal acoustic modes driving combustion instabilities
 - Developed Python image processing scripts using OpenCV to extract flame boundaries from temperature fields and compute mean flame width and RMS fluctuations
 - Implemented POD on temperature field data to extract dominant modes and visualize mode shapes, energy coefficients, and frequency spectra to characterize flame dynamics

RESEARCH PUBLICATIONS

Manuscripts in preparation

1. **Nhan Truong**, Jacob Gamertsfelder, Prashant Khare. “Effect of Embedded Cavities on Fragmentation Behaviors of Droplets in Shock-Laden Flows”. Targeted journal: *Physics of Fluids or Journal Fluid Mechanics*
2. Kamin Manu, **Nhan Truong**, Prashant Khare. “Large Eddy Simulations of Spray and Combustion Processes in a Bluff Body Stabilized High-Speed Combustor”. Targeted journal: *Applications in Energy and Combustion Science*
3. Jeremy Redding, Tripathi Mitansh, Jacob Gamertsfelder, Karmat Siddharth, **Nhan Truong**, Prashant Khare. “High-Temperature Multiphase Dynamics: Shock-Droplet Interaction in Volatile and Metallic Fluids”. Target *Journal of Fluid Mechanics*
4. **Nhan Truong**, Kamin Manu. “Large Eddy Simulations of Recirculation Structures in Isothermal Swirling Coaxial Jet”. Target journal: *Physics of Fluids*

Conference Proceedings/Presentations

1. **Nhan Truong**, Jacob Gamertsfelder, Prashant Khare. “Effect of Embedded Cavities on Fragmentation Behaviors of Droplets in Shock-Laden Flows”. *16th Dayton Engineering Sciences Symposium Conference*, Dayton, U.S.A, October 2024

PROFESSIONAL EXPERIENCE

CFD/ CAE Engineering Intern

May 2023 - Nov 2023

MiSmart | Remote Control UAVs/Drones Design & Manufacturing Start-up

Southeast Asia

- Conducted aerodynamic analysis for over 20 airfoils and 10 remote-controlled aircraft fuselages using ANSYS Fluent, validating aerodynamic performance and ensuring design verification
- Analyzed airfoil performance data using Python libraries, delivering weekly reports to senior leadership on aerodynamics characteristics, supporting performance improvements on VTOL aircraft
- Designed and tested quadcopter UAVs in SolidWorks, achieving a 15% reduction in production costs and a 10% improvement in flight-time durability through iterative ground testing
- Collaborated across departments, including Electrical, AI/ML, Design, and Testing teams to provide aerodynamics data that informed LiDAR object-scanning systems design

TECHNICAL PROJECT

Manufacturing and Testing Engineer

Aug 2022 - Apr 2023

AeroCats | Remote Control Aircraft Design & Manufacturing Club

Cincinnati, Ohio

- Constructed and assembled 4 RC aircraft and UAVs with 16-foot wingspans for national intercollegiate competitions, utilizing laser-cutting techniques and over 1,000 pieces of hardwood, pine, and balsa
- Conducted data analysis on rotational speed and thrust for 6 turbine models of varying lengths, using 3 distinct testing methods to evaluate aircraft electrical engine performance
- Analyzed and compared thrust force data from each turbine to identify the optimal design and calculated maximum payload capacity to ensure readiness for flight testing

EXTRACURRICULAR ACTIVITIES

Programming Chair

May 2025 - Present

American Institute of Aeronautics and Astronautics at University of Cincinnati

Cincinnati, Ohio

- Directing planning and execution of 17+ events across the academic year, including general body meetings, faculty panels, industry sessions, Air Force Museum tour, and GE facilities tour
- Coordinating all logistics including reserving venues, estimating budgets, ordering and picking up catering
- Engaging with professors and professionals to organize speaking engagements and technical workshops
- Coordinating travel for 15+ UC students to attend AIAA SciTech, including conference registration, accommodations, and travel support

HONORS & AWARDS

- Davis Professor R.T. Award for the Best Undergraduate Student in Computational Research Mar 2026
- Best Presentation Award in Combustion at 16th Dayton Engineering Sciences Symposium May 2025
- Undergraduate Research Fellowship, University of Cincinnati Oct 2024
- UC Innovation Challenge Scholarship Jan 2023 - Dec 2024
- Fourth Overall in the Regular Class Competition in SAE Aero Design West May 2023
- University Merit Scholarship, University of Cincinnati Aug 2022 - May 2027
- University Global Scholarship, University of Cincinnati Aug 2022 - May 2027

TECHNICAL SKILLS

- **Programming:** Fortran, Python, C++, Bash, C, MATLAB, LabVIEW, Visual Basic for Application
- **High-Performance Computing:** SLURM, Linux
- **CFD Open Source:** MFC, JAX-Fluids, FluidX3D, ECOGEN, Eilmer4, Gerris
- **Machine Learning:** JAX, PyTorch, Sklearn
- **Simulation Software:** STAR-CCM+, ANSYS Fluent, Converge, XFLR5
- **Data Visualization/ Acquisition:** ParaView, Tecplot 360, Blender, Pov Ray
- **CAD Modeling:** SolidWorks, Fusion 360

REFERENCES

- Dr. Prashant Khare, Ph.D. (Undergraduate supervisor)
Bradley Jones Chair & Associate Professor, Department of Aerospace Engineering & Engineering Mechanics, University of Cincinnati, Cincinnati, Ohio, U.S.A.
Associate Fellow American Institute of Aeronautics and Astronautics (AIAA)
Email: Prashant.Khare@uc.edu; Phone: (513)-556-5248
- Dr. Kamin Manu, Ph.D. (Undergraduate advisor)
Associate Researcher, Department of Aerospace Engineering & Engineering Mechanics, University of Cincinnati, Cincinnati, Ohio, U.S.A.
Email: kaminms@ucmail.uc.edu; Phone: (513)-556-3548